

Danial Kamali

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Education

Michigan State University PHD IN COMPUTER SCIENCE (GPA: 4.0, RESEARCH: VISION & LANGUAGE)	East Lansing, Michigan, USA Aug 2022 - Present
Iran University of Science and Technology B.SC IN COMPUTER ENGINEERING (RESEARCH: FACIAL EXPRESSION RECOGNITION, NLP LEXICAL ANALYSIS)	Tehran, Iran Sep 2016 - March 2021

Experiences

University of California, Los Angeles VISITING GRADUATE STUNDENT	Los Angeles, USA Sep 2025 - Present
• Conducting research on vision-language models with a focus on compositional reasoning.	

Michigan State University GRADUATE RESEARCH ASSISTANT	East Lansing, USA Sep 2022 - Present
• Implemented various deep learning tasks in DomiKnowS (A Library for Integration of Symbolic Domain Knowledge in Deep Learning) for the paper.	
• Researching Compositional Generalization on multi-modal systems improves generalization in unseen structures.	
• Lead a team of researchers on data gathering, labeling, and baseline preparation on the misinformation detection task.	

Qualtrics^{XM} APPLIED SCIENCE INTERN	Seattle, USA Jun 2025 - Aug 2025
• Conducted AI/ML research at the Data Intelligence Center of Excellence (DICE).	
• Revamped the Qualtrics Assist data analytics agent, leading to a 20% improvement in performance.	
• Boosted recall in context retrieval through advanced retrieval and context reduction strategies.	
• Enabled the system to handle a broader range of diverse and compositional question types efficiently.	

Zibal BACKEND TEAM LEADER/FULL-STACK DEVELOPER	Tehran, Iran Sep 2021 - Aug 2022
• Stack: ReactJS, Ant Design, LESS.	
• Managed a team of 4 people with the Agile manifesto and improved the team capacity utilization.	
• Reviewed backend team merge requests on Gitlab and reduced the defection rate.	
• Mentored two interns through 8 weeks of training, one of them joined our team as a junior developer.	

BACKEND DEVELOPER	Oct 2019 - Aug 2021
• Stack: Django, DRF, TensorFlow2, MongoDB, Postgres, Redis, Celery, Docker, Kubernetes	
• Developed data aggregation pipelines and dashboard generation for BI system from NOSQL to SQL-based DB for the product team.	
• Implemented Django-MongoDB generic view for auto dashboard design generation, leveraging reusable patterns to increase developer productivity.	
• Designed CI/CD using Gitlab and kubernetes for front-end and back-end and automated the developemnt and deploy operations.	

IUST Data Mining Lab NATURAL LANGUAGE PROCESSING RESEARCH ASSISTANT	Tehran, Iran Jun 2019 - Dec 2021
• Evaluated Farsi Sentences Tokenization and analyzed current Farsi tokenizers.	
• Mentored undergrad thesis projects and provided feedback on different stages of their project.	

IUST Multi-Agent Systems Lab MACHINE LEARNING RESEARCH ASSISTANT	Tehran, Iran Jun 2019 - Sept 2019
• Researched on Image Caption Generation and customized image captioning in Farsi.	
• Gathered the Farsi image captioning dataset using the coco dataset.	
• Trained the model using the attention mechanism on the mentioned dataset, using TensorFlow.	

Selected Publications

- **Kamali, Danial**, & Kordjamshidi, P. "NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language." *International Conference on Learning Representations*, 2025.
- Nafar, A., **Kamali, Danial**, Karimian, H. R., & Kordjamshidi, P. "An Agentic Framework for Neuro-Symbolic Programming." EACL 2025. Under Review.
- Sinha, S., Premisri, T., **Kamali, Danial**, & Kordjamshidi, P. "Neuro-Symbolic Frameworks: Conceptual Characterization and Empirical Comparative Analysis.", 2025. Under Submission.
- **Kamali, Danial**, Barezi, E. J., & Kordjamshidi, P. "NeSyCoCo: A Neuro-Symbolic Concept Composer for Compositional Generalization." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- **Kamali, Danial**, Romain, J., Liu, H., Peng, M., & Kordjamshidi, P. "Misinformation Detection Using Persuasive Writing Strategies." *Joint International Conference on Computational Linguistics, Language Resources, and Evaluation (COLING)*, 2024.
- **Kamali, Danial**, & Kordjamshidi, P. "Syntax-Guided Transformers: Elevating Compositional Generalization and Grounding in Multimodal Environments." *GenBench: Workshop on Generalization Benchmarking in NLP, Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Skills

Programming	Python, Java Script, C/C++, JAVA, C#
AI Frameworks	Sklearn, Tensorflow, PyTorch , Keras, HuggingFace, Numpy, OpenCV, NLTK, LangGraph , Langchain
Tools	Kubernetes, Docker , MYSQL, MongoDB , Redis, Celery , Kafka
Frameworks and Libraries	Django , DRF, React, Express, Flask, Unity
Language	Farsi: Native English: Professional

Projects

Neural Machine Translation

- Initiated the development of Finglish ol'Saltalneh, an NMT model for Farsi-to-English transliteration, enhancing translation for English forms.
- Curated a dataset of 100,000+ Farsi-English samples, leading data-cleaning efforts to standardize input and improve model training outcomes.
- Devised and applied data augmentation techniques to bolster the dataset, resulting in a 20% BLEU score enhancement for the model.

Facial Expression Recognition

- Developed a convolutional neural network (CNN) that achieved state-of-the-art accuracy on the RAF-DB dataset and competitive results on FER2013.
- Implemented data augmentation and transfer learning techniques to improve model generalizability.
- Implemented statistically aware layers to the network, increasing the interpretability of facial feature representations.

Memotion Analysis: MultiModal Image-Text classification

- Engaged in Memotion Analysis competition on CodaLab, developing a multimodal classification system using NLP and Computer Vision.

Real-time Augmented Reality Sudoku Solver

- Engineered a real-time AR application that captures Sudoku puzzles via video stream, solves them, and overlays the solutions.
- Implemented a custom digit recognition algorithm using neural networks that achieved 95% accuracy on handwritten digits.
- Developed the AR overlay system using OpenCV and Unity, resulting in an interactive and user-friendly application experience.

Awards

2025	Recipient COGS Professional Development award	MSU
2025	Recipient Graduate School Research Enhancement Award	MSU
2022	Recipient Dr. Carl Page Sr Fellowship	MSU
2020	Ranked 1st Annual evaluation of CE department students	IUST
2015	Ranked among the top 0.2% of the candidates National Entrance Exam for Graduate Schools	National
2013, 2014	Qualified for the second phase National Mathematics Olympiad	National
2013	Qualified for the Iran National Team International Mathematics Competition	National
2012	Exceptional Talent Scholarship Entrance Exam	Salam High School

Talks

2025	Compositional Reasoning Under Perceptual Uncertainty UCLA NLP	LA
2025	NeSyCoCo: A Neuro-Symbolic Concept Composer for Compositional Generalization Midwest Speech & Language	Notre Dame
2025	NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language ResearchTrend.AI	Germany
2025	NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language SpaVLE NeurIPS	San Diego

Services & Activities

2024	Student Volunteer EMNLP'24 GenBench Workshop	MSU
2022-2025	Reviewer ACL'23, EMNLP'23, IJACI'24, AAAI'26	MSU
2023	President Iranian Students Association	MSU
2022	Treasurer Iranian Students Association	MSU
2022	Development Team GAMEIN Event	Sharif University

Selected Academic Courses

MSU	IUST
• Deep Learning: 4.0 (top 3) • Computational Foundation of AI: 4.0 (top 1) • Natural Language Processing: 4.0 • Advanced Algorithm Design: 4.0	• Computer Vision: 4.0 • Deep Learning: 4.0 • Computational Intelligence: 4.0 • Natural Language Processing: 4.0